What 9,843 Taiwanese buildings say about rental yield, and about how little of the market any yield can describe

A measurement of Taiwan's rental yield from 實價登錄 sales and leases, 112S4–115S2. First written on 2026-09-29, the day the project was parked; revised since, and § 16 records how.

2.55%	1.73%	6.25%	9,843
median gross yield (building level)	median net yield (after four stated costs)	of 157,407 buildings could be priced at all	matched buildings (143,297 sale–lease pairs)

Paper version	v1.3 · 2026-11-13
Payload version	v2.13 · built 2026-11-13
Data window	112S4 → 115S2 (11 quarters, Q4 2023 → Q2 2026)
Source snapshot SHA-256	bb889edb4fd9589e41edc2a58d168fa3…
Code	github.com/Stanley0210/taiwan-rental-yield
First written	2026-09-29

Every figure in this report is read from housing_data.json at build time. Where the report describes numbers that were published earlier and later withdrawn, it says so; those figures come from the project handoff, not the payload.

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Executive summary

The one-sentence version. Take a real sale from Taiwan's 實價登錄 registry, find a real lease in the same building within four quarters, and the rental yield falls out with no modelling at all. Done nationally, that gives a median gross yield of **2.55%** (95% CI 2.52–2.56) and a net yield of **1.73%** (95% CI 1.72–1.75) — below the 2.315% the five big banks charge on a new mortgage. But the more important number is the denominator: those medians describe **9,843** of **157,407** buildings that sold in the window — **6.25%**, about one in 16.

The project's thesis was that **the denominator is the point**: every transaction-based yield published about Taiwan describes a small, young, tower-heavy, investor-owned slice of the market, and nobody says how small. This project measured it.

Ten findings that survived every audit

- **Coverage is structural.** 6.25% of buildings with a sale can be priced. 72.0 percentage points are lost because no lease exists in the registry at all; the project's own quality filters cost 21.7pp. Even with every filter removed, the ceiling is 24.20%.
- **Towers are over-represented roughly sevenfold relative to walk-ups.** Match rates: towers (住宅大樓) 10.98%, mid-rises (華廈) 4.73%, walk-ups (公寓) 1.54%. Towers are 40.3% of buildings that sold but 70.8% of those that can be priced.
- **The priced slice is newer and dearer.** Matched buildings were completed around 民國96 against 84 for the rest, and cost 8.2% more per 坪. And 42.7% of Taiwan's dwellings (1–3 storey 透天) are outside the analysis by construction.
- **Re-weighting the visible slice to look like the market raises the yield** from 2.55% to 2.81% (+0.26pp). The bias is published, not silently corrected.
- **Under stated assumptions, the yield of all 157,407 buildings lies in [1.90%, 4.26%].** With no assumptions it is the entire quality band, and no amount of better cleaning can change that, because a median can only be bounded once more than half the population is seen. **That assumption is now tested, and it only half holds:** the cells absorb 77% of the age gap between matched and unmatched buildings and make the size gap worse (§ 6). The interval is a stated-assumption interval, not a validated one.
- **Social housing distorts raw registry rents.** 82.1% of whole-unit leases are subsidised 社會住宅 lets, renting 28.0% below market per 坪. Anyone computing a yield off raw 實價登錄 rents is mostly measuring the subsidy.
- **Yield differences between districts are mostly price.** The correlation between a district's price per 坪 and its yield is -0.757; rents vary far less than prices do.
- **Inside one building, rent barely rises with floor (+0.10% per floor) while price rises five times faster (+0.50%).** Furnished units rent 4.8% higher than unfurnished ones in the same building. These are within-building comparisons, so building quality is held constant.
- **Pre-sale (預售屋) is only 5.4% dearer** (CI 4.2–7.0) than genuinely completed new homes sold in the same contract quarter. That is not the widely quoted ~30%. The 30% came from a comparison group that was 90% pre-sale contracts itself.
- **Un-itemised parking is not free.** Leases with a space whose rent was never itemised pay the same total rent per m^2 as itemised ones (ratio 1.032, CI 1.004–1.079). Imputing the space rent moved the headline from 2.68% to 2.55%.

What did not survive

The project published several things that later checks withdrew, and it keeps the withdrawals on the record:

- **The founding hypothesis — walk-ups out-yield towers — is closed as unanswerable.** The gap exists in the matched data (+0.506pp, CI 0.403–0.647), but matched towers are 18% smaller than towers that sold and could not be matched. Two differently skewed samples cannot settle a population question. The proposed mechanism (management fees) was falsified outright.
- **Distance to the MRT** was published, retracted, restored and finally withdrawn. It turned on whether 輕軌 (light rail) counts as a station, and Kaohsiung showed no gradient.
- **The "~30% pre-sale premium"** was built on a contaminated comparison group (see §12).
- **"One building in 1,086"** divided buildings by registered 建號 (≈ units). That is a unit error, and it has been withdrawn.
- **A price index** made the yield-vs-gap slope worse, not better, so it is switched off.

> **In plain English**
>
> A landlord who bought a typical priceable Taiwanese flat in this window earns about 2.5% a year in rent before costs and about 1.7% after them, which is barely more than a one-year bank deposit (1.70%) and less than the 2.31% mortgage rate. But "typical" here means a newer flat in a tower, because that is the only kind the public data can price. For about 15 in every 16 buildings the data simply cannot say, and this report shows exactly where each of those buildings was lost.

1 Origin and constraints

Quants.tw began as a Taiwan-stock chip-flow product. On 2026-08-30 the same repository was restarted as a **Taiwan rental-yield analysis tool**: same infrastructure, different subject. The stock product was frozen at the time, so no live users were affected. The first version covered Taipei and New Taipei (2,549 matched buildings, gross 2.43%). The scope constraint was lifted on 2026-08-31 and everything since covers all 22 counties. Over the following four weeks the payload went from v1.0 to v2.8, the test suite from ~780 to 919, and the method was audited by independent agents in ten dimensions.

The engine is, in effect, underwriting: pull a real sale, pair it with a real lease in the same building, and run the net-operating-income arithmetic. Three standing rules came from Stanley and never changed:

- **This was the second start.** A third pivot would itself be the signal, so the rule was to ship badly and iterate, not redesign.
- **Stay in the data-and-analysis lane.** The site analyses housing data. It does not opine on housing policy.
- **Yields come from matched pairs, never district averages.** Scope went national; this rule did not move.

The legal line. 不動產估價師法 §14 reserves 估價 (valuation) for licensed appraisers, with fines of NT$30k–150k. The site therefore says "transactions in this building cleared at X% on these assumptions" and never "this property is worth NT$Y" or "this is a good buy". Every building panel ends with a line saying the figure is a registered transaction price, not a valuation.

Status when parked (2026-09-29). The code lives on branch feat/housing and tag housing-parked-2026-09-29 on the quants remote. The quants.tw domain has returned Railway's "Application not found" since late August, because the Railway account that hosted it lapsed; the code was never the cause. No reader ever saw the housing site live. The stock site was restored on main the same day this report was written.

2 The data

Primary source: 內政部 不動產成交案件實際資訊資料供應系統 (實價登錄). It is free, needs no key, and is published under 政府資料開放授權條款-第1版. The combined season zip carries sales (a), pre-sales (b) and leases (c) for every county.

	raw rows	after cleaning	kept
Sales (買賣)	970,942	395,426	40.73%
Leases (租賃)	432,649	66,705	15.42%
Pre-sales (預售屋)	—	179,252 dwellings	4,580 建案

Why the window starts at 112S4 (Q4 2023). Six lease columns first appear that quarter: 出租型態, 有無管理員, 租賃期間, 有無電梯, 附屬設備 and 租賃住宅服務. Every quality filter depends on them. Using older seasons would silently blend subsidised social housing and room-shares into the early period and manufacture a fake time trend. The usable window is therefore 11 quarters, which is too short for any claim about cycles.

The filters (each can be switched off, so the honesty ladder in §7 walks the same code path):

- Residential types only: 住宅大樓 (11+ floors, lift), 華廈 (≤10 floors, lift), 公寓 (≤5 floors, no lift). 1–3 storey 透天 are excluded by construction.
- Social housing (社會住宅包租轉租 / 代管) excluded, because those rents are subsidised.
- Whole-unit lets only (整棟(戶)出租, 獨立套房). This drops room-shares, floor lets and blank types, whose area field is not comparable to a whole-unit sale.
- Related-party sales (親友 / 親屬 / 關係人) excluded, because they are not market prices.
- Parking subtracted from both sides; un-itemised space rent imputed (§ 13).
- Sale and lease within ±4 quarters; yields outside 0.3–12% rejected as filing errors.

Addresses. A building key is derived from the 門牌 (house number) by a normaliser tested on 150 labelled pairs. Only 0.01% of records produce no key. Keys were corrected in v2.8 for two assumptions the registry falsifies: the variant character 市 (U+5DFF) for 市, and unit designators containing 號. Coordinates come from per-county open 門牌 point files, because Taiwan has no national open geocoder: 8,804 of 9,843 buildings (89.44%) were placed, with 100% district agreement. Coordinates feed only the map, never a yield.

Context sources, each hashed in the payload: 內政部 statistics (stock by floor and age, 法拍 volumes, low-use housing, price index, unsold new homes), 中央銀行 (deposit, rediscount and mortgage rates), 主計總處 household income, 台灣金服 金拍 auction results, and OpenStreetMap (basemap and 264 metro/light-rail stations).

3 The method

The unit is the building, not the pair. Pairs concentrate heavily in mega-towers: the median building has 2 pairs, the largest has 6,988, and the top ten buildings hold 17.1% of all pairs. A pair-level median (2.79%) overweights exactly the buildings that are least typical. Every headline figure is a median of per-building medians.

Matching a sale to a lease on the same property is not a new idea. Bracke (2015) does it for Central London from a letting agency's own book: 1,922 properties both sold and let within six months. What differs here is the source and what it lets you say — a public national registry rather than one firm's clients, and a registry whose non-coverage is itself measurable, which an agency's book cannot be.

Intervals. All confidence intervals come from one two-stage bootstrap: resample buildings, then pairs within each building. Pairs inside a building are not independent draws, and it is the number of buildings, not the number of pairs, that sets the precision (Cameron et al. 2008) — which is also why districts carry a floor of 25 buildings before they are reported at all. The browser recomputes intervals with a seeded RNG so they do not move between page loads.

Net yield. Four costs are deducted, each a live slider on the site. The JavaScript implementation is a twin of the Python one and is tested to agree on every building.

Cost	Default	Range	Source
Holding tax	0.15 % of price / yr	0.05–0.4	Peng et al. (2007) 0.089-0.130%, Wang & Tseng (2013) 0.066%, Li ~0.12% -- all self-use. 囤房稅2.0 raised the non-self-use house-tax leg from 1.2% to 2.0-4.8%, so the default sits above the measured range.

Cost	Default	Range	Source
Management fee	140 NT$/坪/月	0–300	Taipei general residential runs NT$100-180 per ping (New Taipei 90-120). Applied only to buildings whose lease records report on-site staff. WARNING: those two rates are the only ones with a published source, and the site now covers 22 counties -- the other 20 are all charged New Taipei's rate for want of a per-county statistic. What that assumption is worth is measured below. Separately, about four in ten leases in staffed buildings state in 備註 that the rent includes the fee and five in ten say nothing; whether the fee should be charged at all is bounded under "what the lease itself says" (advanced mode).
Vacancy	5 %	0–20	No Taiwan rental-vacancy statistic exists. The often-quoted 8.77% is the 111H2 reading of the electricity-derived LOW-USE rate -- a different population, and one that moves every half-year (this page also publishes 111H1 9.52%, 114H1 10.43% and 114H2 9.44%). 5% is an assumption, not a measurement; what can be measured about its SHAPE is in the vacancy section.
Repairs	6 % of rent	0–25	No public source. Charged as a share of rent and scaled by age (x(1+age/50)), because a flat rate biases the walk-up-vs-tower comparison in one direction.

Reproducibility. The payload carries a per-season SHA-256 manifest of the source zips, hashes of every reference CSV and the 金拍 cache, the build commit and a version number. bin/rebuild-housing.sh rebuilds it and diffs it structurally against the committed hash.

4 The headline yield

2.55%
gross, 95% CI 2.52–2.56

1.73%
net of tax, management, vacancy, repairs
95% CI 1.72–1.75

2.79%
the same data, pair-weighted (the wrong unit)

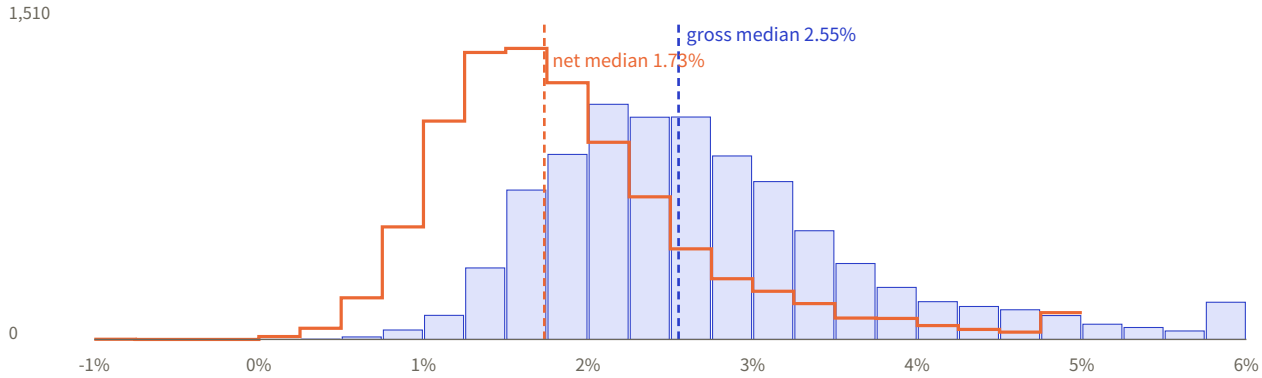


Figure 4.1 — Distribution of building-level yields. Bars: gross; line: net. The end bars are pile-ups of everything beyond them (the histogram clips), not counts of their own interval. The horizontal distance between the two medians (0.82pp) is the cost model.

874 buildings (8.9%) net less than 1% a year, and 1 nets less than zero. Against the reference rates published beside it:

Reference	Rate	Period	What it is
Net yield, this site	1.73%	112S4–115S2	median across matched buildings
1-yr time deposit	1.70%	Nov 2025	five big banks, identical at all five
Rediscount rate	2.00%	since 22 Mar 2024	policy rate, not investable
New mortgage (CBC)	2.315%	Sep 2026	five big banks, new house-purchase loans
New mortgage (MOI)	2.69%	114Q4	all new loans nationally

A buyer financing with a mortgage in this window pays more in interest than the median priceable flat returns in net rent. The yield is below the cost of the money that bought it. That is an arithmetic statement, not a verdict on whether the price is wrong: under the user-cost identity (Himmelberg et al. 2005) a low rent-price ratio can be a rational response to low real rates and expected appreciation, and nothing measured here settles which it is.

The unit-size objection

A pair here is a sale and a lease in the same building, not the same unit, and the headline divides total rent by total price. Nothing makes the two units the same size: a leased studio set against a sold three-bedroom reads as a low yield, and the reverse as a high one. No edition before v1.3 said so, and it is the first thing an appraiser would ask. Bracke (2015) avoids the problem by matching the same property. This registry carries no unit identifier, so the same building is as close as it gets — but what the mismatch is worth can be measured, three ways:

Reading	median gross	95% CI	buildings	pairs
Headline — total rent ÷ total price	2.55%	2.52–2.56	9,843	143,297
Per 坪 — rent per 坪 ÷ price per 坪, pair by pair	2.53%	2.50–2.54	9,842	143,139
Size-matched — only pairs whose areas are within 0.8–1.25×	2.55%	2.52–2.57	8,556	88,605

Nationally the headline holds. The size-matched reading is the closest this registry gets to a same-unit design, and it assumes nothing about how rent scales with floor area. It returns 2.55% on 87% of the buildings, a shift of +0.00pp; per 坪 moves it by -0.02pp. The reason is symmetry, not luck. Across pairs the median ratio of sold to leased area is 1.00; 20% of pairs have the sold unit larger than the band allows and 19% the leased one, so the mismatches cancel in a median.

Locally it holds less well, and the comparison below is where that matters. Where a city's mismatches are not symmetric the three readings separate, so every city figure in the next table is given all three ways and each gap is quoted as a range.

Against other published estimates

Taiwan already has a rental-yield figure in circulation, and it is worth asking where it comes from. Chang and Chen (2018) give a gross rent multiplier of 64 for Taiwan and a yield of 1.57%; the stated source of that table is Global Property Guide, a commercial listings aggregator. Global Property Guide (2026) now publishes 2.26% nationally, computed as median asking rent × 12 ÷ median list price over apartments in five cities. So the peer-reviewed Taiwan number and the consumer-facing one are the same number, and neither is built on transaction data. Two differences from the measurement in this report follow. An asking price is not a sale price. And a ratio of two medians taken over different properties is not a median of per-property ratios — they are different statistics, and only the second one is a yield any individual building could have earned.

The comparison city by city, which has not been made before:

City	headline	95% CI	per 坪	size-matched	buildings	GPG (June 2026)	gap, three readings
新北市	2.26%	2.23–2.29	2.29%	2.27%	2,583	2.19%	+0.07 to +0.10pp
高雄市	2.91%	2.86–2.93	2.82%	2.87%	1,980	2.57%	+0.25 to +0.34pp
臺北市	2.08%	2.03–2.12	2.09%	2.10%	1,616	1.97%	+0.11 to +0.13pp
桃園市	2.56%	2.48–2.63	2.58%	2.57%	949	2.44%	+0.12 to +0.14pp
臺南市	3.07%	2.94–3.14	2.92%	3.00%	792	2.12%	+0.80 to +0.95pp

The registered measure is higher in all five under all three readings: by +0.07 to +0.10pp in 新北市, and by +0.80 to +0.95pp in 臺南市. The two sources agree closely in the north and diverge going south, and that ordering does not depend on the reading. **The size of the southern gap does.** v1.1 and v1.2 quoted 臺南市 at +0.95pp from the headline reading alone. That is the top of the range and not the middle of it: once unit size is controlled the gap is +0.80pp per 坪 and +0.88pp on size-matched pairs. §16 records the correction.

Earlier editions also hedged that a municipality-wide median covers "a great deal of non-urban area" that a listings site does not. That is checkable, because 臺南 and 高雄 each merged with their county in 2010 and the old city limits are still district boundaries. In 臺南 the pre-merger city proper yields 3.20% (2.97% per 坪, 477 buildings) against 2.85% (2.78%) in the former county: the wider footprint lowers the municipal figure, so it hides part of the gap instead of creating it. In 高雄 it runs the other way and is small — 2.88% (2.78%) in the city proper against 3.04% (3.00%). What stays unlike-for-like is the source and the window: asking prices against registered ones, and a 2026 snapshot against eleven quarters.

Taipei is the one city with three independent estimates, and they agree. 楊博宇 (2018) estimate a Taipei gross rent multiplier of about 50 from 591 asking rents against 實價登錄 sales for 2013–2016, which inverts to a yield near 2.0% — an arithmetic inversion of their figure, not one they state. Global Property Guide gives 1.97% for 2026. This report gives 2.08% (2.09% per 坪, 2.10% size-matched). Three methods, three windows, a spread of about a tenth of a point.

Set against the bounds in §6 rather than against the point estimate: the current listings figure and the 2013–2016 multiplier both fall inside the resemblance interval, and the 1.57% in circulation falls below it — though that figure describes an earlier period than this window, so it is not a contradiction so much as a reminder that a yield without a date and a denominator is not a measurement.

This table is typed, not derived. It is the only one in the report that is. These figures live on other people's pages and in other people's PDFs; no payload can produce them and no test can check them. They were read from the sources on 2026-11-03 and the sources are in §19, so a reader can re-check them the way they can re-check everything else here.

5 Coverage: the central finding

Of 157,407 buildings with a recorded residential sale in the window, 9,843 can be priced. The waterfall says where every other building went, and whose fault each step is.

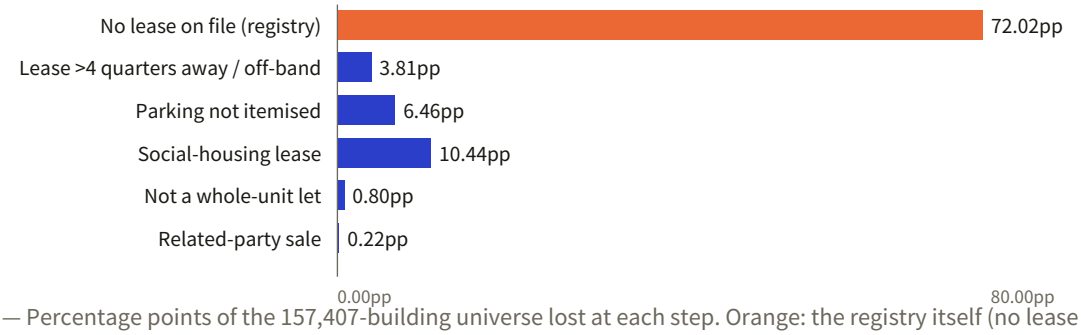


Figure 5.1 — Percentage points of the 157,407-building universe lost at each step. Orange: the registry itself (no lease on file), 72.0pp. Blue: this project's quality filters, 21.7pp combined. The law-side share of the loss is 77%.

"No lease on file" is not the same as "unregistered lease", and the reason is in the law rather than in the data. Only two duties to register a lease exist. A broker must register the leases it brokers or agents (不動產經紀業管理條例 §24-1), and a master-lease business must register its subleases (租賃住宅市場發展及管理條例 §34), each within thirty days. A landlord and a tenant who sign with each other directly have no duty at all, so their lease never reaches the registry — and most of these buildings are owner-occupied anyway. No amount of cleaning can recover a lease that was never filed. With every one of this project's filters removed, the ceiling is **24.20%**.

Which buildings survive

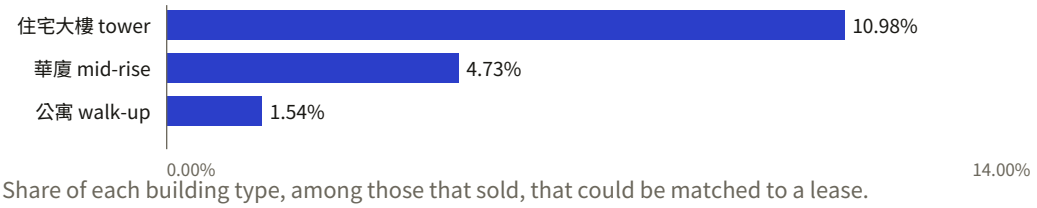


Figure 5.2 — Share of each building type, among those that sold, that could be matched to a lease.

Type	sold (棟)	matched	match rate	share of sold	share of matched	gross	net
住宅大樓	63,435	6,964	10.98%	40.30%	70.80%	2.44%	1.63%
華廈	44,795	2,120	4.73%	28.50%	21.50%	2.96%	2.08%
公寓	49,177	759	1.54%	31.20%	7.70%	2.94%	2.27%

Against the whole housing stock

The government's house-tax roll counts 9,550,294 dwellings (宅). **42.7% are 1–3 storey buildings**, mostly 透天厝, which the residential filter excludes by definition, so no result here speaks for them. Within the 4+ storey stock the site does describe, towers are 42.3% of dwellings but 70.8% of priced buildings, and walk-ups are 38.0% against 7.7%. The stock counts dwellings and the site counts buildings; because a tower holds many more dwellings than a walk-up, this comparison understates the skew. The two are set side by side and never divided.

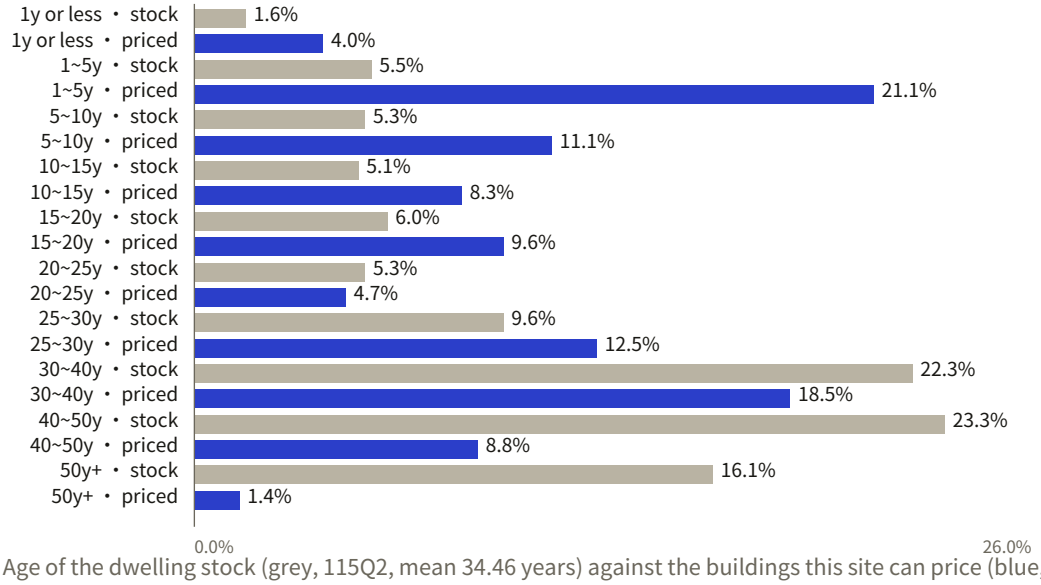


Figure 5.3 — Age of the dwelling stock (grey, 115Q2, mean 34.46 years) against the buildings this site can price (blue, median 19 years). The priced slice is concentrated in buildings under ten years old; over-40s are nearly absent.

The selection effect in one table

This is the classical selection problem (Heckman 1979; Manski 1989), and housing measurement has met it before: an index built only from houses that sell twice is biased because selling twice is not random (Gatzlaff and Haurin 1997). The usual repair is to model the probability of being observed and correct for it. That repair is unavailable here, and §6 explains why: being observed depends on whether a lease was filed through a licensed agency, which is an institutional fact about the registry rather than a behaviour of the building, and the registry publishes nothing that predicts it.

	matched	unmatched	difference
Buildings	9,843	147,564	

	matched	unmatched	difference
Price per 坪 (median)	325,896	301,142	+8.2%
Completed (民國, median)	96	84	+12 years newer
Area (坪, median)	31.6	33.9	-6.9%

This comparison shipped at a third of its size for two versions. It read the cleaned sale frame, which had already dropped the cheaper stock (un-itemised parking, related-party), so it compared against 100,800 buildings instead of all of them and printed a 2.7% price gap. On the full universe the gap is 8.2%. A test now pins the unmatched count to universe minus matched.

Re-weighting the visible slice

Post-stratifying the matched buildings to the universe's mix moves the gross yield from 2.55% to 2.74% (price decile alone), 2.69% (type alone) and **2.81%** (both, 95% CI 2.78–2.85); net moves to 1.98% (CI 1.95–2.01). The interval excludes the unweighted 2.55%, so the +0.26pp shift is not a rounding artefact. The cheapest price decile is under-represented (0.34× its share) and the dearest over-represented (2.14×). Because cheaper buildings yield more, the visible slice understates the yield of the buildings it cannot see — if they resemble their price-decile neighbours. §6 shows why that "if" cannot be removed.

Affordability, as a side result

Measured as median price over average income per earner (主計總處, 2024), a home costs 14.4 years of income nationally and 13.6 for the priceable slice. Taipei is the outlier at 22.7 years; 嘉義市 is the most affordable at 7.7. The priced slice is dearer per 坪 but smaller, so its total price is lower.

6 Bounding the 94% nobody can see

Every number above is a median of the buildings the site can match. Dropping the other 147,564 silently assumes they look like the ones on file. Partial identification refuses that assumption and bounds them instead. Inside each cell (type, then type × price decile, then × age band), matched buildings enter at their observed yields and the unmatched share enters as a mass at the cell's assumed floor or ceiling. The weighted median of that mixture is a sharp bound for the assumption named on each rung. This is the standard construction (Manski 1989; Horowitz and Manski 1995; Manski 2003): put the unobserved mass at the extremes of its support, and what comes out is an interval that the data cannot narrow rather than a point the data cannot support.

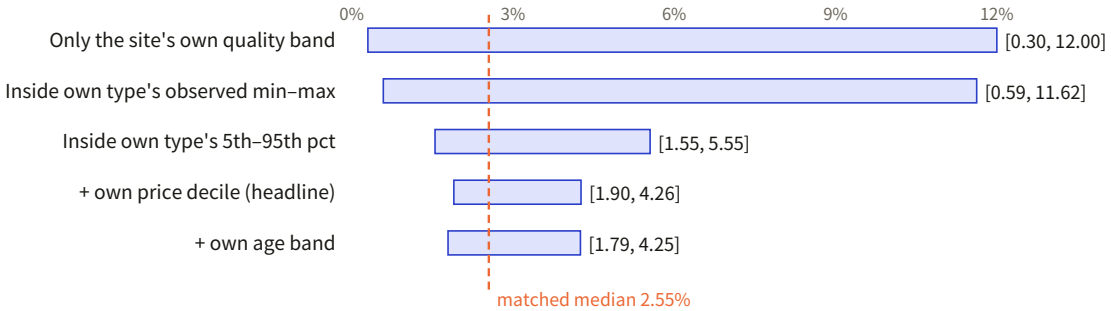


Figure 6.1 — Where the median gross yield of all buildings with a sale can lie, under each rung's assumption. Each rung is sharp for its assumption, not merely conservative.

The headline rung puts the median of all buildings in **[1.90%, 4.26%]** gross and [1.08%, 3.22%] net. Three things constrain how to read it:

- **The trim is a resemblance assumption.** "Inside the 5th–95th percentile" forbids any unmatched building from sitting in its own type's top 5%. It buys 7.04pp of narrowing, the largest step, so the untrimmed rung stays beside it.
- **The age rung fails and is published failing.** 75 of 144 cells hold too few matched buildings, 21,395 buildings fall back to the weaker assumption, and the interval widens by 0.10pp. Conditioning on more does not always identify more.
- **The residual selection is unsigned.** Matched buildings are 8.2% dearer per 坪, which suggests unmatched buildings yield more. But matching requires both a sale and a lease, which selects liquid investor stock, which suggests they yield less. Both directions are published, and neither is chosen.

> **The quotable result**
>
> A median's worst-case interval only leaves the support band once more than 50% of the population is observed. This site observes 6.25%; with every filter removed the registry allows 24.20%. No amount of better cleaning moves the first rung. That is a fact about the registry, not a disclaimer about the site.

Applied to walk-ups alone (1.54% matchable), the interval is [1.55%, 5.78%], 4.23pp wide. That is wide enough to swallow the entire walk-up/tower gap of §9.

Is the resemblance assumption true?

Every rung above the support rung rests on one idea: that an unmatched building resembles the matched buildings of its own cell. Editions up to v1.1 stated that, laddered it and labelled it honestly — and never tested it, which made it the only load-bearing claim in this report that was asserted rather than measured. It is testable, on the two things the cells do not condition on: how old a building is, and how big it is. If the cells do what the rung claims, the gap between matched and unmatched buildings should close inside them.

Reported as a standardised difference — the mean gap over the pooled standard deviation — which is scale-free, with $|d| < 0.1$ the conventional threshold for balanced. Cells need 25 matched buildings to count, so a handful of buildings cannot outvote a decile of them.

	age (built year)	area (坪)
Unconditional d, whole universe	+0.520	-0.219
Within-cell |d|, size-weighted	0.119	0.531
Within-cell signed d	+0.104	-0.531
Absorbed by the cells	77.1%	-143.0%
Cells not balanced	14 of 30	26 of 30

Age: mostly absorbed. Across the whole universe matched buildings are +0.52 standard deviations newer. Inside the cells that falls to 0.119, so the conditioning removes 77% of it. What survives is small but it is not noise: the signed mean is +0.104, the same direction as the raw gap, and 14 of 30 cells are still outside the balanced band. Matched buildings are newer than their own cell-mates.

Area: made worse, not better. Unconditionally matched buildings are 0.22 standard deviations smaller. Inside the cells that does not shrink — it grows to 0.531, 143% in the wrong direction, with 26 of 30 cells unbalanced and every one of the worst six carrying the same sign. The mechanism is not mysterious: the cells are cut on price per 坪, and 坪 is the denominator of that. Banding on a ratio strips out the price channel that was partly offsetting the size channel, so within a band the size gap expresses itself more freely than it did across the whole universe.

Worst cells by |d| on area: 華廈 decile 7 (-0.95), 華廈 decile 6 (-0.85), 華廈 decile 3 (-0.83), 公寓 decile 4 (-0.80).

What this does to the interval

So the headline interval is a **stated-assumption** interval, not a validated one, and on one measurable dimension the assumption is not merely unverified but contradicted. Nothing about the arithmetic changes — the bound is still sharp for the assumption named on its rung — but a reader should treat the narrowing from the quality band as resting on a condition the data only half supports.

It cannot be repaired by conditioning on size as well. The age rung already shows what happens: adding a third dimension puts 75 of 144 cells below the minimum and sends 21,395 buildings back to the weaker assumption, which is why it widens the interval by 0.10pp. At 6.25% coverage the data runs out before the conditioning does. That is the same wall as the rest of §6, met from a different direction.

7 What each cleaning step is worth

The honesty ladder re-runs the headline with filters added one at a time, all through the same code path as the headline. Each rung is read per building with a 95% CI; the pair-level figure is shown beside it to make the unit problem visible.

Rung	per building	95% CI	per pair	buildings	pairs
Every lease record, unfiltered	2.27%	2.26–2.28	2.60%	38,053	1,010,047
Parking subtracted, both sides	2.51%	2.50–2.52	2.88%	27,888	754,727
Social housing removed	2.67%	2.64–2.68	2.72%	11,455	183,648
Whole-unit lets only (room-shares, floor lets and blanks removed)	2.71%	2.69–2.73	2.92%	10,190	152,287
Related-party sales removed	2.68%	2.65–2.69	2.89%	9,843	143,303
Per building, not per pair	2.68%	2.65–2.69	2.89%	9,843	143,303
Un-itemised space rent imputed and subtracted	2.55%	2.52–2.57	2.79%	9,843	143,297
Parking-bundled transactions dropped whole (the old treatment)	2.88%	2.84–2.90	3.11%	5,807	87,664

Because filter order matters, each filter's contribution is also computed as a **Shapley value** (Shorrocks 2013) over all 2^3 combinations; the three sum exactly to the total move of +0.171pp per building. Social housing: **+0.179pp** (the largest, and the sign the subsidy predicts). Whole-unit only: +0.029pp. Related-party: -0.038pp.

The Simpson reversal that forced this

An earlier ladder showed removing social housing lowering the yield, which is the wrong sign for a population renting 28% below market. The filter was correct; two presentation faults stacked:

- **Mixed units.** A building holding any social lease carries 32.9 pairs against 7.7 for one with none, so a pair-level median weights social buildings 14.1:1 and a building-level one 3.3:1.
- **Ordering.** Room-shares yield ~1.27% (their area field describes the building, not the let), are 14.5% of market leases and 0.0% of social ones. Dropping social housing first raises the room-share weight.

Let type	social	market	social − market
獨立套房	3.08%	3.12%	-0.04pp
整棟(戶)出租	2.61%	2.85%	-0.24pp
分租套房	0.90%	1.27%	-0.37pp

Within every let type the social lease yields less, yet the aggregate moved the other way. The page's own sentence claiming the ladder measured "what each cleaning decision is worth" was retracted: a sequential ladder cannot do that, and the Shapley decomposition can.

The social-housing rent gap

Social-housing leases rent at NT$627 per 坪 per month against NT$871 for market leases (-28.0%), and they are 82.1% of whole-unit registry leases (~58% of all leases). The share is above 90% in 桃園, 苗栗 and 基隆. Excluding them was the single largest filter decision in the project, and it is the one most raw-data yield estimates skip.

8 Geography

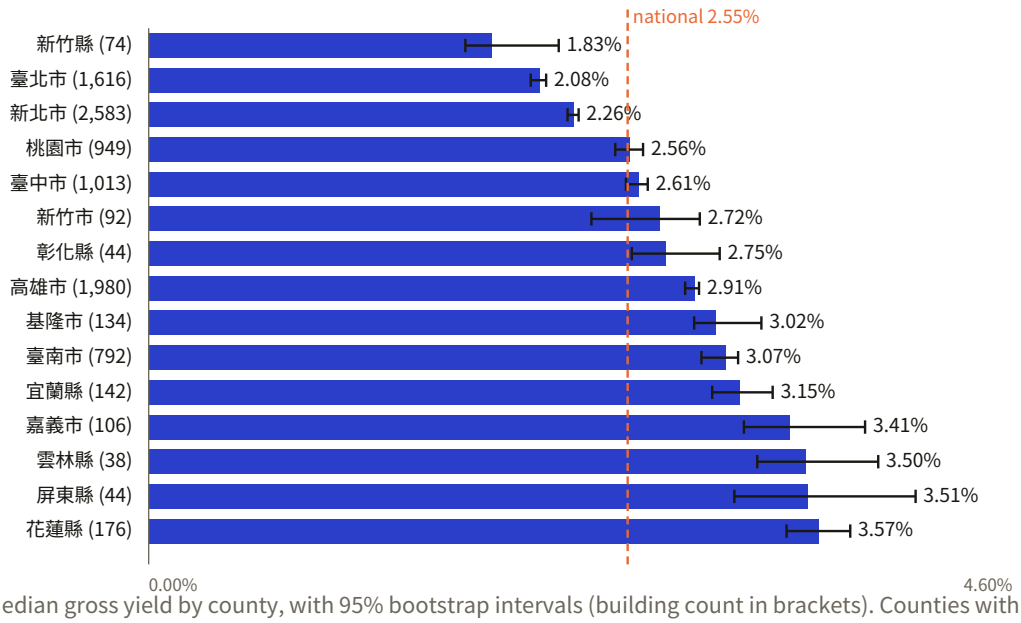


Figure 8.1 — Median gross yield by county, with 95% bootstrap intervals (building count in brackets). Counties with fewer than 25 matched buildings are omitted. 金門 and 連江 have no clean leases at all; that is a registry fact.

Most adjacent counties cannot be told apart statistically (80 of 80 adjacent district intervals overlap), but the two extremes genuinely differ. Taipei is the lowest-yielding large city because it is the most expensive: across districts the correlation between price per 坪 and yield is **-0.757** (over the 81 published districts). Rents vary much less than prices do.

	District	gross	net	buildings	price / 坪
lowest	新竹縣 竹北市	1.67%	1.11%	53	555,815
	臺北市 南港區	1.79%	1.22%	74	958,577
	臺北市 大安區	1.84%	1.32%	240	1,140,969
	新北市 新店區	1.86%	1.28%	182	688,275
	新北市 土城區	1.91%	1.34%	112	600,889
	臺北市 中正區	1.91%	1.43%	105	1,047,685
highest	屏東縣 屏東市	3.77%	2.53%	26	163,572

District	gross	net	buildings	price / 坪
臺南市 南區	3.60%	2.79%	39	195,302
花蓮縣 花蓮市	3.60%	2.62%	143	198,761
高雄市 新興區	3.51%	2.35%	77	249,788
臺南市 中西區	3.49%	2.70%	76	236,318
嘉義市 嘉義市	3.41%	2.19%	106	181,366

Within vs between. Over 3,440 buildings with at least 4 pairs, the typical spread of yields inside one building (IQR 0.533pp) is 0.64× the spread between district medians (0.837pp). In the two-city version this ratio was 2.46× the other way, "one building varies more than a whole city". That conclusion inverted when scope went national and was rewritten to follow the data in either direction. Even so, two flats in one building differ by more than half as much as two districts do, which is why a single district average hides a lot.

Vacancy break-even. The highest-yielding district with enough data, 屏東縣屏東市 (3.77%), would need a vacancy rate of 26.5% to fall to the national district median — against a default of 5% and a measured county low-use rate far below that. High yields in cheap districts are not explained away by empty flats.

9 Walk-ups vs towers: the founding hypothesis

The project started from one idea: older 公寓 walk-ups, cheap per 坪 and without management fees, should out-yield towers. It went through four verdicts.

Stage	Verdict	Why
Two cities (Aug)	dead	gross gap 0.041pp; ~81% of the net gap was the project's own cost assumption
National (Sep 8)	alive	gross gap +0.374pp, CI excluding zero; the "dead" label had been a hard-coded string
Controls (Sep 11)	contested	+0.120pp within price decile, +0.637pp within district: both cannot be the effect
v2.2 (Sep 20)	unanswerable	tower-side representativeness fails; fee mechanism falsified

The gap is real in the matched data. Walk-ups 2.94% vs towers 2.44%: +0.506pp gross (CI 0.403–0.647, 4.3× the minimum detectable difference) and +0.645pp net. It holds in the same sign across most control families:

Control family	cells	walk-ups covered	median gap	same sign	binding
price decile	9	735	+0.315pp	89%	yes
district	28	548	+0.728pp	82%	yes
county	6	688	+1.090pp	100%	yes
age band	3	607	+0.288pp	100%	yes
size band	4	759	+0.331pp	100%	yes
has manager	2	759	+0.711pp	100%	yes
district × price decile	40	328	+0.344pp	70%	no
district × age band	0	0	—	—	no

Why it cannot be settled. Matched walk-ups look like walk-ups in general (largest median difference 3.8%). Matched towers do not: they are 34.8 坪 against 42.5 for towers that sold and never matched (-18.0%). Two differently skewed samples cannot answer a population question, and the pre-stated rule classifies that as unanswerable selection. The age distributions barely overlap either: walk-up median 45

years vs tower 14, and zero district × age cells reach 25 buildings of each.

What is settled: the proposed mechanism is false. A manager is recorded for 97.3% of towers and 6.1% of walk-ups, so the fee is nearly a type indicator. But in buildings with no manager on either side (713 walk-ups vs 188 towers), where the modelled fee is exactly zero, the gap is still +0.467pp (CI 0.208–0.572). The gap does not need the fee. The project's founding explanation died on the project's own data.

What data would answer it

A complete population of residential leases (owner-let walk-ups never enter the registry, which only receives filings from licensed firms), or building-level house-tax micro-data by type × age × floor (released to institutions only). Neither is available to this project.

10 Inside one building

Comparing units within the same building holds location, age, construction and amenities constant by construction. These are the most strongly identified results in the project.

Comparison	effect	buildings	basis
Furnished vs unfurnished rent	+4.8%	3,210	NT$37/坪/month on 782
Rent per floor	+0.10%	3,777	NT$0.78/坪 per floor
Price per floor	+0.503%	22,299	NT$1,632/坪 per floor

Buyers pay about five times more per extra floor than tenants do, so higher floors yield less: over twenty floors that is roughly 0.40pp of yield. The view is capitalised into the price, not the rent. The two slopes come from different building sets (3,777 with leases on 2+ floors, 22,299 with sales on 2+ floors), and the site names both counts.

11 Transit: a finding withdrawn four times over

Distance to the nearest station looked like the project's second finding: yield rising from about 2.1% near a station to 2.9% beyond 2.5 km, monotone across seven bins. It is now **withdrawn**, and the path there is the most instructive story in the project.

Version	What was published	What was wrong
v1.2	national gradient, "price falls faster than rent"	only Taipei's 131 stations were loaded: 高雄 buildings were measured ~275 km to Taipei
v1.3	retraction on a 臺北市-only panel	Taipei City has 4 buildings beyond 2.5 km; the "null" was an empty group
v1.3b	restored on 北北桃, then 新北 alone (+0.53pp)	heavy rail only; 淡海輕軌 serves exactly the far group
v1.4	withdrawn	with light rail the 新北 gap is ~0; 高雄, the first independent city, shows none

Recomputed for this report from the per-building distances in the payload (medians of building gross yields, ≤500 m vs >2 km):

County	Stations counted	≤500 m	>2 km	far − near
新北市	heavy rail	2.10% (n=641)	2.46% (n=798)	+0.36pp
新北市	incl. light rail	2.25% (n=1053)	2.33% (n=348)	+0.08pp

County	Stations counted	≤500 m	>2 km	far − near
高雄市	heavy rail	2.97% (n=516)	2.82% (n=396)	-0.16pp
高雄市	incl. light rail	2.93% (n=758)	2.81% (n=329)	-0.13pp

The rule this produced. A result that flips on a judgement call (is 輕軌 a station?) is a finding about the judgement, not the subject. Where such a call exists, compute both and show both. And a null from a group of four is not evidence of absence: check the size of the comparison group before believing a null.

12 Pre-sale: the 30% premium that was not one

The registry's pre-sale files hold 179,252 dwellings in 4,580 projects across 226 districts in the window (median NT$427,956 per 坪). A pre-sale flat has no lease, so no yield can ever be computed for one; that half of the result never moved. The price comparison was published wrong twice:

- **Attempt 1** compared pre-sale against "completed buildings under two years old", found +30.2%, and attributed it to the delivery wait.
- **Attempt 2** retracted the mechanism (a placebo between two completed groups fell almost as fast) but kept the +30.2%.
- **Attempt 3** found the group itself was wrong. 交易年月日 in the sale file is the contract date, not the handover date, so "under two years old" selected 186,550 rows of which **168,426 (90.3%) were contracts signed before the building existed**, median age at contract −2.17 years. The published number was pre-sale against pre-sale.

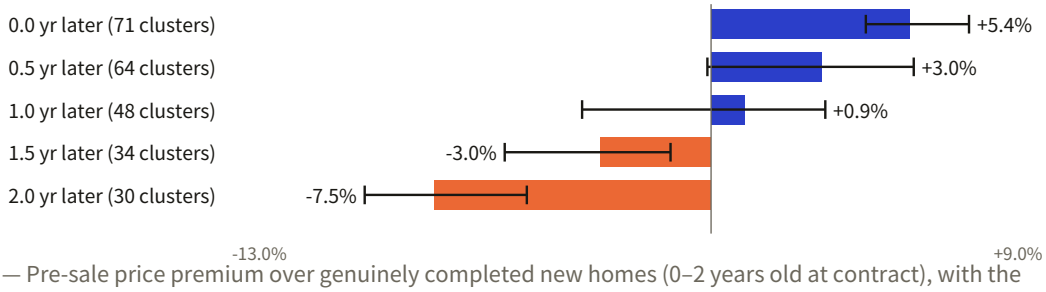


Figure 12.1 — Pre-sale price premium over genuinely completed new homes (0–2 years old at contract), with the comparison pushed later in time. Intervals are bootstrapped over district × type clusters, not cells, because cells within a cluster are not independent.

Only the contract-quarter gap stands: **+5.4%** (CI 4.2–7.0), stable at minimum cell sizes of 5, 25 and 100. It reaches 63,361 dwellings (35.3%) in 63 of 226 districts; the rest have too few genuinely completed new builds to compare against. Whether the later gaps are time or composition cannot be separated (separable = false).

13 Parking: a cleaning assumption tested and found false

32.00% of clean leases (21,331) bundle a parking space whose rent was never itemised. Until v2.0 the pipeline assumed that meant the space came free and subtracted only its area. The test: in buildings holding both kinds of bundled lease, compare total rent per m².

	value
Predicted ratio if the space were free	0.885
Predicted ratio if its rent were inside the dwelling rent	1.000
Measured (n = 191 buildings)	1.0317 [1.004, 1.079]
Verdict	assumption fails

The assumption fails, so the dwelling rent was overstated on those leases. The typical itemised space rents for NT$3,000/month (11.5% of the lease). Three treatments were compared:

Treatment	gross	net	buildings
Impute space rent (headline since v2.1)	2.55%	1.73%	9,843
Area-only subtraction (v1.0–v2.0 headline)	2.68%	1.85%	9,843
Drop bundled leases entirely	2.73%	1.88%	6,840

Dropping them looks attractive but flatters the result: the 30.5% of buildings that would vanish are newer (民國107 vs 88) and 85.4% towers. The rule that decided the verdict was written after an early n=192 look, and the page says so rather than calling it pre-registered.

Parking as an asset

Treated as its own investment, a space bought and let at market prices yields **2.00%** gross (CI 1.95–2.12, 275 buildings, median price NT$1,800,000, rent NT$3,000/month), below the flats above them. Mechanical spaces (坡道機械) yield more than ramp-level spaces (+0.82pp on the wider sample); this is labelled a hypothesis, because its mechanism is not measured.

14 The cost model, and how much each guess matters

Net yield rests on four assumptions, and two of them (vacancy, repairs) have no Taiwanese source at all. The project's answer was to measure how much each guess moves the result, not to pretend it had a source.

Management fee

Only Taipei and New Taipei have published fee surveys, and the other 20 counties are charged New Taipei's NT$105/坪. 48.1% of buildings are charged a fee at all (those whose leases record on-site staff), and 5,644 sit in counties with no fee statistic. Halving the non-Taipei rate moves net to 1.89%; zeroing it moves it to 2.04%. Separately, 39.7% of leases in staffed buildings state in 備註 that the rent already includes the fee. Charging only buildings without that flag gives 1.88% (+0.14pp).

Vacancy

Taiwan publishes no rental-vacancy statistic, so the level stays an assumption (5%). The shape can be measured from 內政部's low-use (electricity-based) housing table crossed by county and age, recovered from the Internet Archive:

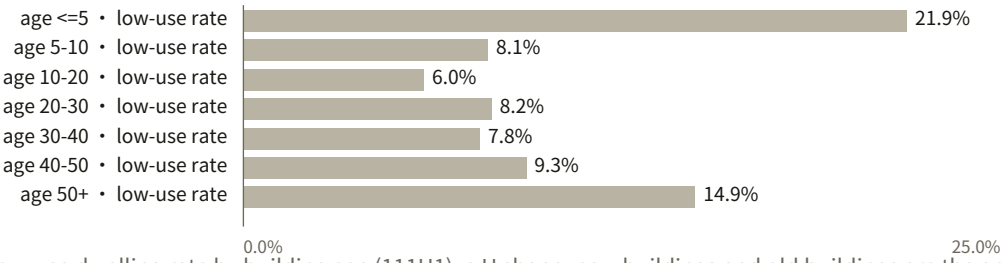


Figure 14.1 — Low-use dwelling rate by building age (111H1), a U shape: new buildings and old buildings are the emptiest.

Age band	low-use rate	share of priced	share of stock
<=5	21.91%	25.02%	7.11%
5-10	8.08%	11.08%	5.26%
10-20	5.97%	17.80%	11.09%
20-30	8.20%	17.17%	14.92%
30-40	7.82%	18.44%	22.28%
40-50	9.35%	8.77%	23.28%
50+	14.92%	1.41%	16.05%

The priced slice sits on the left arm of the U, where a flat vacancy rate flatters the yield most: weighted by its own composition, its expected low-use rate is **11.53%** (CI 11.40–11.66) against 9.52% nationally. The county-only leg points the other way (9.62% vs 10.43%), because the sample is metropolitan; both are published. Low-use is not rental vacancy (its numerator includes second homes and homes for sale), so it never replaces the headline assumption:

Case	net	vs headline	walk-up − tower (net)
a flat 5% for every building (the headline)	1.73%	+0.000pp	+0.645pp
the 5% level, shaped by the low-use rate	1.73%	-0.006pp	+0.700pp
charging the low-use rate outright	1.55%	-0.182pp	+0.742pp

Repairs are charged at 6% of rent × (1 + age/50). A flat rate would bias walk-up vs tower in one direction: the walk-up gap is 0.506pp gross, 0.645pp with age-scaled repairs and 0.757pp with a flat rate. **Holding tax** defaults to 0.15% of price, above the published self-use estimates, because 囤房稅2.0 raised the non-self-use rate.

15 Market context

Transaction volume halved inside the window

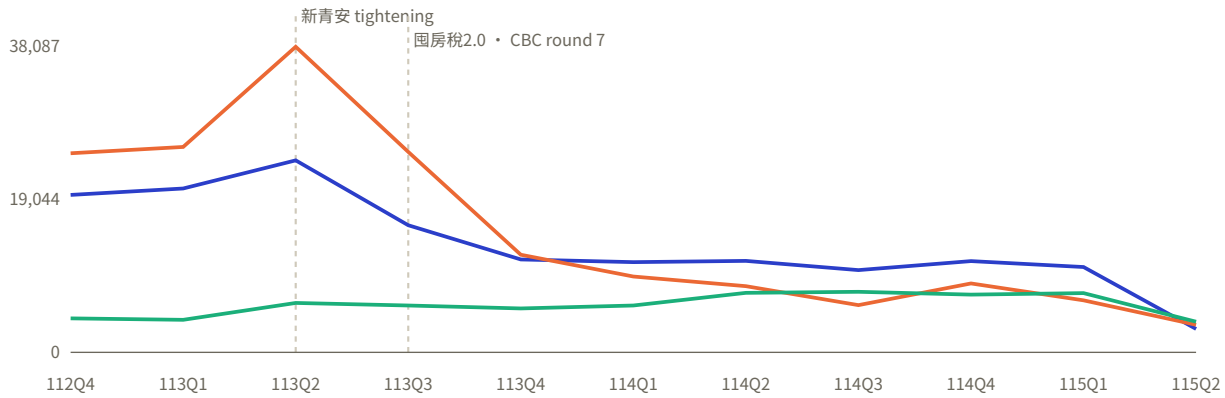


Figure 15.1 — Clean registry rows per contract quarter, nationally. Blue: existing homes; orange: pre-sale; green: leases. 115Q2 is incomplete (late filings).

Existing-home sales fell 51.6% from their 113Q2 peak (23,929) to 113Q4 (11,578); pre-sale fell 84.5% by 114Q3. Leases rose from 6,158 to 7,384 over the same span. Three policies landed in the same two quarters, so the page describes the drop and does not attribute it.

The priced slice is dearer than the official average

County	official mean 元/坪	site median 元/坪	gap	buildings
新北市	442,000	488,011	+10.4%	2,583
高雄市	271,200	295,945	+9.1%	1,980
臺北市	763,300	890,773	+16.7%	1,616
臺中市	305,700	321,798	+5.3%	1,013
桃園市	292,800	326,548	+11.5%	949
臺南市	255,800	259,079	+1.3%	792
花蓮縣	206,200	198,923	-3.5%	176
宜蘭縣	221,800	210,314	-5.2%	142
基隆市	204,600	218,450	+6.8%	134
嘉義市	220,000	181,366	-17.6%	106

Against 內政部's 115Q1 contract averages, the site's matched buildings are dearer in every large city except 花蓮, 宜蘭 and 嘉義市. This is the "visible slice is dearer" finding confirmed from outside the registry.

Uncorrected drift. The official price index rose 5.62% over 9 quarters (0.61%/quarter). At the median sale–lease gap of 2 quarters, a typical pair carries about 1.22% of price drift. The project's own index made things worse — the yield-vs-gap slope steepened from 0.029 to 0.063 at ±4 quarters — so it is off, and the drift is disclosed instead.

Forced sales

Court auctions (法拍) cleared 5,202 lots worth NT$625億 in the window. The bank-collateral channel (金拍, 台灣金服) the project could scrape had 1,765 lots in the same window (33.9% of the court count). These are parallel channels, not a subset. Among 金拍 lots, 34.0% sold above reserve (first round 32.9%). No area field exists, so no auction discount is computed. 法拍 itself is unreachable: 司法院's system refuses connections.

Also published: 112,501 completed-but-unsold new homes at 114Q4; national low-use rate 9.44% (114H2), from 6.39% in Taipei to 10.92% in Tainan.

16 Corrections register

The project's design principle was to refuse flattering versions of itself: when a result looked clean, the next step was to try to break it. These are the published claims that later checks changed. Every one was true, or at least defensible, when it was written.

Claim	First published	Now	Caught by
Headline gross yield	2.43% (2 cities) → 2.68%	2.55%	scope; parking audit
Hero coverage sentence	"only 6.22% have both on file"	27.8% have both; 6.25% survive our filters	independent audit
Match rates by type	towers 14.09%, walk-ups 1.69%	10.98% / 1.54%	denominator moved to universe
Selection effect (price/坪)	+30.5% → +2.7%	+8.2%	site audit (wrong frame)
Walk-up hypothesis	dead → alive	unanswerable; mechanism falsified	tower-side check
Within vs between spread	within 2.46× between	within 0.64× between	national rescope
Transit gradient	survives	withdrawn	light-rail definition; 高雄
Pre-sale premium	+30.2%	+5.4%	contract-date contamination
Pre-sale net-of-time	0.9%	not separable	second audit
Priceable share of all buildings	1 in 1,086	withdrawn (棟 ÷ 建號)	own MOI audit
Social-housing ladder rung	−0.165pp	Shapley +0.179pp	unit/order analysis
Un-itemised parking	space came free	rent inside dwelling rent	within-building test
Net histogram & sliders	"sliders move this chart"	they did not; fixed	4-agent audit
Auction registry share	13.8% of buildings	14.0% of lots (unit named)	audit
Room-share filter label	"room-shares"	also blanks and floor lets (5,964)	data-cleaning audit
臺南 gap vs listings (§ 4)	+0.95pp	+0.80pp per 坪, +0.88pp size-matched	pre-send stress test (v1.3)
City footprint (§ 4)	"a great deal of non-urban area"	measured: 臺南's city proper yields more than its former county	pre-send stress test (v1.3)
Lease-filing rule (§ 5)	asserted	cited to the two statutes	pre-send stress test (v1.3)

Across the full ten-dimension site audit: 41 findings raised, 35 confirmed and fixed, 6 rejected on verification. Two of the six rejections were false alarms raised by the project itself (a rounding floor mistaken for a bug, and an income "sign flip" that compared two different populations).

Editions of this paper

Numbers get a corrections register; the write-up itself needs one too, or a version number says nothing about what moved.

Edition	Date	What changed
v1.3	2026-11-13	Three things found by stress-testing the paper before sending it to anyone. A pair is a sale and a lease in the same building, not the same unit, and no earlier edition said so: section 4 now measures what the size mismatch is worth, per 坪 and on size-matched pairs. The national headline survives it; the city comparison does not survive it unchanged, and the gap against the listings figure is restated on a size-normalised basis, which is smaller in the south. The hedge about city footprints is replaced by a measurement. And the registry's lease-filing rule, asserted for three editions, is now cited to the two statutes it comes from. The new-mortgage reference rate is read from a pinned central-bank file instead of typed, which also brings it from May to September 2026.
v1.2	2026-11-04	Tests the resemblance assumption that every rung above the support rung rests on, and publishes the result, which is that it only half holds: the cells absorb 77% of the age gap between matched and unmatched buildings and make the size gap WORSE. No number moved -- the interval is the same and was always labelled assumption-dependent -- but its standing changed, from an assumption nobody had checked to one the data contradicts on a measurable dimension. Section 6 says so.
v1.1	2026-11-03	Adds section 19, the references, and with it the first comparison against published estimates (section 4) -- including that the Taiwan yield in circulation traces to a listings aggregator rather than to transaction data. Section 18 rewritten: the project is published, so the items saying nobody had read it and the repositories were private had stopped being true. No measured figure changed.
v1.0	2026-11-01	First citable edition, written when the project was parked.

17 Bug classes and working rules

The same few mistakes recurred in different clothes. These rules are the most transferable output of the project, and they apply to any future data work, stocks included.

- **Derive, don't type.** Nine published sentences went stale when scope went national, because numbers update themselves and prose does not. Every figure in the copy is now computed from the payload.
- **Branch, don't assert.** Where a claim has a direction (survives or not, wider or narrower), the copy picks its branch from the data.
- **Assert the derivation, not the answer.** A test that checks verdict == "dead" protects a stale conclusion; a test that checks the verdict follows the interval does not.
- **Check the unit of both numbers before the arithmetic.** Pair vs building, lot vs building, 棟 vs 宅 vs 建號, windowed vs unwindowed: this was the most frequent error class.
- **A rate needs its numerator and denominator from the same frame.**
- **Compute both sides of a comparison before writing its verdict.** The walk-up verdict flipped the moment the tower side existed.
- **Establish which population a number describes before calling it wrong.**
- **Read the rendered page, in both languages.** Placeholder-order bugs (a count printed as a percentage) pass every arity test and are visible only on screen.
- **A tolerance calibrated on one dataset is a constant pretending to be a measurement.**
- **A closure over the thing you are resampling is a resampling that does not happen.**
- **A hard-coded colour is only right in one theme.**
- **Size multi-agent audits to finish.** 10-agent runs died on session limits; 3–5-agent runs completed and every completion found real bugs. When verifiers die, verify by hand.

18 Open items and how to restart

v1.0 wrote this section about a parked project, and its first three items were that no reader had ever seen the work, that both repositories were private, and that the page carried noindex. None of those is true now. A register that goes on asserting them is worse than no register, so this edition splits the list.

Closed since v1.0

- **It is public.** Write-up, payload, pipeline and page are at github.com/Stanley0210/taiwan-rental-yield, served at stanley0210.github.io/taiwan-rental-yield.
- **"Check my denominator" is a real invitation now.** The payload ships with its SHA-256 and a provenance block hashing every source archive and reference file the build read.
- **It is citable**, with a Zenodo concept DOI that always resolves to the newest version, a CITATION.cff, and the citation in the README.
- **It exists in Traditional Chinese** as well as English, built from the same payload by a second generator, with a test pinning that both editions state the same headline figures.
- **The stale ceiling is fixed.** v1.0 caught its own typed constant — partial_id.ceiling_pct read 24.0% while the waterfall said 24.20% — and named the one-line repair. Payload v2.9 reads it from the waterfall, so the two can no longer disagree.
- **The net yield and the reweighted median now carry intervals.** Both were published as bare point estimates beside a gross figure that had a CI.
- **Dependencies are pinned and a clean install is tested.** v1.2 pinned requirements.txt and then printed "dependencies are unpinned" in this very list, which is the staleness this section exists to catch, committed inside the section itself. requirements-dev.txt now also carries pytest, which lived nowhere despite being the command the README leads with, and CI installs from those pins on a machine that has never seen the project, verifies the payload hash, runs the suite and rebuilds all four documents.
- **The mortgage rate is read, not typed.** v1.2 concluded it had to stay a transcription because the CBC publishes 五大銀行新承做放款金額與利率 only as a PDF. That was wrong: the listing page has no data file, but each month's release page carries a spreadsheet with the whole series back to 1994, and nobody had opened a release page. It is now pinned and hashed beside its two siblings, and the figure is read from it. The pinned file gives exactly the 2.322% that used to be typed for May 2026, so the old number was never wrong — only unverifiable, and four months stale. It now reads Sep 2026.

Found while preparing this edition

A sort that was not a total order

A rebuild from identical inputs and identical code did not reproduce the published payload. Thirty fields differed, every one of them in the 內政部 low-use table, and all of them the same seven rows in reverse order. The cause was a sort: sorting on the period column with sort_values defaults to a non-stable quicksort, and once the frame is filtered to the latest period every key is tied — so the surviving row order was a property of the installed pandas, not of the data.

On its own that is a reproducibility hole. But both editions of this paper read that array by position — the last row as the national rate, row 5 as Taipei, row 1 as Tainan — so a rebuild under a different pandas printed Kaohsiung's rate as the national one and swapped the two cities. Three confidently labelled wrong numbers, with the PDF building cleanly and every test passing. Both reports now look rows up by county code and never by position, the row order is pinned regardless, and a test shuffles the rows and asserts the rendered figures do not move.

Still open

- **quants.tw itself is still unbound on Railway**, so the interactive version of this work — the sliders, the per-district pages — runs nowhere. The static page carries the findings; it does not carry the tool. The runbook is docs/RAILWAY_GO_LIVE.md on the housing branch.
- **The mortgage file cannot refresh itself.** It has no stable URL — each month is a new release page — so a newer month means someone pinning a newer file. A test fails when the pinned month falls a year behind the data window; until then the staleness is visible in the file's manifest entry rather than hidden in a constant.
- **Per-building URLs** (/housing/b/<key>) were never built, because there was no live site to link to.
- **The rental-vacancy level** remains an assumption; only its shape is measured (§ 14).

Data that could be added next

A scouted and critic-checked menu lives in docs/HOUSING_NEXT_DATA.md. Its "do first" items (volume series, lease-stated management fees, parking audit) were all shipped in v1.9–v2.0. The remaining useful sources are 法拍 at item level (blocked by 司法院), building-level tax micro-data (institutions only) and a national 門牌 geocoder (TGOS, institutions only). All three are blocked by eligibility, not effort.

How to bring it back

Step	Command / action
Check out the code	git checkout feat/housing (or the tag housing-parked-2026-09-29)
Run locally	SECRET_KEY=devtestkey QUANTS_SUBJECT=housing venv/bin/python3 app.py
Rebuild the payload	venv/bin/python3 src/underwrite.py (needs the 207 MB season zips in data/lvr/)
Verify reproduction	bin/rebuild-housing.sh
Tests	venv/bin/python3 -m pytest tests/ -q (265 in this repository; 919 was the whole source site)
Read first	docs/HOUSING_HANDOFF.md §0, §2, §11, §8

The raw inputs (season zips, address files, 金拍 cache) are gitignored and live only on the local machine under data/lvr, data/geo and data/fasc. They are hashed in the payload, so a fresh download can be checked against the published build.

19 References

Each work is here because a claim in this report leans on it, and the line under each entry says which. A reading list of what the subject ought to contain, rather than what the paper actually used, would be decoration.

Partial identification — the frame § 6 works inside

Horowitz, J. L., and C. F. Manski (1995). **Identification and robustness with contaminated and corrupted data**. Econometrica 63(2), 281–302.

The sharp-bound construction section 6 uses: put the unobserved mass at the extremes of its support and the parameter is bounded, not point-identified.

Manski, C. F. (1989). **Anatomy of the selection problem**. Journal of Human Resources 24(3), 343–360.

States the problem this project has: when the outcome is observed selectively, the sampling process does not identify the population quantity, and no amount of data cleaning changes that.

Manski, C. F. (2003). **Partial Identification of Probability Distributions**. Springer Series in Statistics. Springer, New York.

Book-length treatment of prediction with missing data; the source for why a median's worst-case interval is finite only once more than half the population is observed.

Selection in transaction data — why § 5 is the finding and not a caveat

Gatzlaff, D. H., and D. R. Haurin (1997). **Sample selection bias and repeat-sales index estimates**. Journal of Real Estate Finance and Economics 14(1–2), 33–50.

The same structure one step removed: an index built only from houses that sell twice is biased because selling twice is not random. Here it is leasing and selling, and the surviving share is far smaller.

Heckman, J. J. (1979). **Sample selection bias as a specification error**. Econometrica 47(1), 153–162.

The correction this project deliberately does not run: it needs a model of why a lease is filed, and section 5 shows the filing rule is institutional rather than behavioural.

Price and rent measured on the same property

Bracke, P. (2015). **House prices and rents: microevidence from a matched data set in Central London**. Real Estate Economics 43(2), 403–431.

The closest methodological sibling: rent-price ratios from properties both sold and let, matched within six months. Agency data, one city, 1,922 properties; this project's analogue is a public registry, a whole country, and a published denominator.

Himmelberg, C., C. Mayer, and T. Sinai (2005). **Assessing high house prices: bubbles, fundamentals and misperceptions**. Journal of Economic Perspectives 19(4), 67–92.

Why a gross yield is not a verdict: the user-cost identity holds that a low rent-price ratio can be a rational response to low real rates and expected growth. Section 14's net yield is a cost model, not a user cost, and does not settle that question.

Taiwan

Chang, C.-O. (張金鶚), and S.-M. Chen (2018). **Dilemma of housing demand in Taiwan**. International Real Estate Review 21(3), 397–418.

Carries the Taiwan rental-yield figure this paper is measured against. Its Table 5 gives a rent multiplier of 64 and a yield of 1.57%, sourced to Global Property Guide (2018) -- a listings aggregator, not transaction data. That provenance is the gap this project fills.

楊博宇 (Yang, P.-Y.) (2018). **臺北市房地產預期增值為何高漲？——以租金乘數探討投資價值與消費價值的差異**. 碩士論文, 國立政治大學地政學系 (master's thesis, Department of Land Economics, National Chengchi University); advisers 張金鶚, 江穎慧.

Taipei gross rent multiplier of about 50 from 591 asking rents against 實價登錄 sales, 2013–2016. The closest published Taipei number built on the same registry, and it brackets this paper's Taipei median.

The law the registry's coverage rests on

不動產經紀業管理條例. **第 24 條之 1（租賃案件成交資訊之申報登錄）**. 全國法規資料庫 law.moj.gov.tw, pcode D0060066; text in force, read 2026-11-11.

「經營仲介業務者，對於居間或代理成交之租賃案件，應於簽訂租賃契約書之日起三十日內，向直轄市、縣（市）主管機關申報登錄成交案件實際資訊。」

The first of the only two duties to register a lease: a broker must register the leases it brokers or agents, within thirty days. A lease with no broker is outside it.

租賃住宅市場發展及管理條例. **第 34 條第 2 項（包租業轉租案件之申報登錄）**. 全國法規資料庫 law.moj.gov.tw, pcode D0060125; text in force, read 2026-11-11.

「包租業之租賃住宅轉租案件，應於簽訂轉租契約之日起三十日內，向直轄市、縣（市）主管機關申報登錄成交案件實際資訊。」

The second: a master-lease business must register its subleases. Together with the broker duty this is the whole of the registry's lease intake, which is why a building with no lease on file is the rule rather than a data-quality failure.

Method: attribution, and intervals under clustering

Cameron, A. C., J. B. Gelbach, and D. L. Miller (2008). **Bootstrap-based improvements for inference with clustered errors**. Review of Economics and Statistics 90(3), 414–427.

Pairs within a building are clustered, and the cluster count is what limits precision. The two-stage bootstrap resamples buildings first for that reason; this is also the warning that few clusters make intervals optimistic, which is why district intervals carry an n floor.

Shorrocks, A. F. (2013). **Decomposition procedures for distributional analysis: a unified framework based on the Shapley value**. Journal of Economic Inequality 11(1), 99–126.

The reason section 7 reports Shapley values rather than a filter sequence: the Shapley decomposition is the order-independent one, and this ladder's order demonstrably matters.

Compared against — data, not literature

Global Property Guide (2026). **Gross rental yields in Taiwan: Taipei and 4 other cities**. globalpropertyguide.com, last updated June 2026; read 2026-11-03. Asking rents and list prices from a local listings platform, apartments of 1–5 bedrooms in five cities; gross yield computed as median monthly rent × 12 ÷ median purchase price.

The figure in general circulation, and the one Chang and Chen (2018) report. Section 4 sets it beside the registry measurement city by city.

Appendix A — Provenance

Payload v2.13, built 2026-11-13 from commit bae72e03f077+dirty. Source snapshot digest bb889edb4fd9589e41edc2a58d168fa3bffd3ef513466d94a1c557720e0653c6.

Season	SHA-256	bytes
112S4	39fe69d8dc89ec905a4427ed3b5abad03e2b0b048fe37257018be67b1bfd6491	16,714,295
113S1	7c010c79421225c5b96bb0607f4d3b40ec99cf5155861658978419551 9ae7863	16,498,147
113S2	e247d0f20365b790685b5e63a2197d40200e0f436ceb14f3ee88e16ecd095972	19,127,935
113S3	87ffdebbeb59c51b1bb9cbaddeeedac27fbecd57e9024e53615b5ae33b80e5e0	20,978,409
113S4	4bb63c1858d5bc6f821280a7d7dfa81b6daefa610e5dc47cb7f4fb53e330b913	16,051,231
114S1	c541074f34f4c5ecde81e4926823d664a9eb9d09c2b020ba7179604c88bdeb2a	13,718,336
114S2	9562aa1e268bb13698a3118bfaae366edf4499745a637456a953584e5ed2a23c	14,868,369
114S3	a3b3b51c7cdc9b9790a51a78f232fb3875ca5e1ff2f12a32644109dd4ba962bc	15,018,542
114S4	68315c006fa367834eaacf032ceccb22a114fc8798d87b8dac23cf5eb64030ab	13,872,186
115S1	8d0bfbaea4fe56d7aa9876b8d587d51d6f69f3419ba1d4e5bab59ec198fe9fe2	14,252,348
115S2	6b4c148bad106a235dddf214001d605ff9ffa45ac50561f7125c6d7fa795d4da	14,705,972

Reference file	SHA-256 (first 24)	bytes
data/fasc/*.json	ddcdc6bf07166f0859443324…	4,958,158
data/ref/moi/building_stock.csv	19b8e3a0d148ede90f7a4b6d…	1,046,973
data/ref/moi/court_auction_amt.csv	0e5fc68588bc8bd213428a43…	44,428
data/ref/moi/court_auction_n.csv	758f37ab7680a01ac1800455…	34,411
data/ref/moi/housing_stock.csv	93b4757b24ba0cb1e47e27de…	42,683
data/ref/moi/low_use.csv	76181c3c509a1a3e97906706…	3,411
data/ref/moi/low_use_22.csv	20a50814112b1bb66b948fe1…	9,323
data/ref/moi/low_use_by_age.csv	85de19fc65d3b5e4b9f463b4…	8,557
data/ref/moi/mortgage_rate_new.csv	e59e0a4eb70d7ac0e1a5b2cc…	39,233
data/ref/moi/price_index.csv	94adcd497add01c3b7955ef7…	8,617
data/ref/moi/rent_ping.csv	66b354929e7df56bf3136d03…	28,764
data/ref/moi/resi_auction_n.csv	53f88f39a42a5a82a2be23d5…	683,801
data/ref/moi/resi_tax_roll.csv	bf0059d0a14475c02f65af2a…	1,045,168
data/ref/moi/sale_price_ping.csv	beba2fd4232013c14b69a12f…	1,567,640
data/ref/moi/sale_transfers.csv	5c02bb07930b96a2262301c2…	35,053
data/ref/moi/stock_by_age.csv	65d1fbe6a94e2639080ba39c…	3,145,761
data/ref/moi/stock_by_floor.csv	3d0d24d08ce67594eb226bc0…	1,924,035
data/ref/moi/stock_by_size.csv	6828deacd1879576aebe352c…	2,657,819
data/ref/moi/unsold_new.csv	e3e87621edc5cce22f527b8e…	3,588
data/ref/moi/xfer_by_age.csv	5a0ff2840f7bfa47018e2b2b…	2,164,622
data/ref/moi/xfer_by_floor.csv	aede74a2e306563764e9d27a…	1,614,568
data/ref/moi/xfer_by_size.csv	177000ca0169e4ef51dce6bc…	1,855,540
data/ref/cbc/A13Rate.csv	61081da8d1be0065a8d59354…	260,563
data/ref/cbc/five_bank_new_loans.ods	433f83cac11daaa03cddcf80…	55,099
data/ref/cbc/webF1.csv	2b07898ec156f16c5f523265…	1,456
data/ref/dgbas_income.csv	70b93974ceb45e7e5af89036…	5,669

Appendix B — Glossary

Term	Meaning
實價登錄	Taiwan's actual-price registration system for property transactions and leases.
坪 (ping)	3.3058 m^2, the standard unit for Taiwanese floor area.
住宅大樓 / 華廈 / 公寓	Tower (11+ floors, lift) / mid-rise (≤10 floors, lift) / walk-up (≤5 floors, no lift).
透天厝	1–3 storey single-family townhouse; 42.7% of dwellings and outside this analysis.
社會住宅包租代管	Government-subsidised sublet and managed social-housing leases.
預售屋	Pre-sale: a flat sold off-plan, before the building exists.
法拍 / 金拍	Court-ordered auction / bank-collateral auction via 台灣金服. Different channels.
棟 / 宅 / 建號	Building / dwelling / registered building number (roughly one per unit). Never divided by each other.
Gross yield	12 × monthly rent ÷ price, parking removed from both.
Net yield	Gross after holding tax, management fee, vacancy and age-scaled repairs.
Matched pair	A sale and a whole-unit market lease in the same building within ±4 quarters.
Coverage	Matched buildings ÷ all buildings with a residential sale in the window.
Partial identification	Bounding a quantity for an unobserved population under stated assumptions instead of assuming it resembles the observed one.
Shapley value	A filter's average contribution over every order in which filters could be applied.
民國	ROC calendar; year + 1911 = Gregorian (民國115 = 2026). Seasons such as 115S2 = Apr–Jun 2026.